

Program: Diploma in Food Processing Technology	
Course Code: 6422D	Course Title: Non thermal Processing Techniques
Semester: 6	Credits: 4
Course Category: Open elective	
Periods per week: 4(L:4 T:0 P: 0)	Periods per semester: 60

Course Objectives:

1. To impart knowledge on various non thermal processing technologies in food industry
2. To learn in detail about various membrane technologies applied in the food industry
3. To have an in depth understanding of hurdle technology

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of processing techniques	Food Processing technology

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the principles, equipment, processes, and applications of High Pressure Processing (HPP) and Pulsed Electric Field (PEF) in food process engineering.	15	Understanding
CO2	Explain in detail about various membrane technology applied in processing industry.	15	Understanding
CO3	Describe the process of extrusion for producing food items with novel characteristic features	15	Understanding
CO4	Explain the concept and applications of hurdle technology and image processing technology in food processing industries	15	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the principles, equipment, processes, and applications of High Pressure Processing (HPP) and Pulsed Electric Field (PEF) in food process engineering		
M1.01	Explain the fundamental concepts of food process engineering	5	Understanding
M1.02	State the principle, equipment, process, and application of HPP	5	Understanding
M1.03	Explain the principle, equipment, process, and application of pulsed electric field technology	5	Understanding
Contents: Introduction to food process engineering, Principle, equipment, process, and application of HPP, Principle, equipment, process, and application of pulsed electric field			
CO2	Explain in detail about various membrane technology applied in processing industry.		
M2.01	Discuss membrane technologies in food processing	4	Understanding
M2.02	State principle and applications of ultrafiltration and nanofiltration	5	Remembering
M2.03	Outline the principle and applications of reverse osmosis	5	Understanding
	Series Test –I	1	

Contents: Overview of Membrane technologies in food processing, principal and application of membrane technologies- Ultra filtration (UF), Nano filtration(NF) , Reverse Osmosis (RO)			
CO3	Describe the process of extrusion for producing food items with novel characteristic features		
M3.01	Explain food extrusion technology	5	Understanding
M3.02	Classify the types of extruders	5	Understanding
M3.03	Describe the applications of extrusion cooking in food industry	5	Understanding
Contents: Food extrusion technology – theory, Equipment and its functioning, Types of extruders, Effect of extrusion on foods, Application of extrusion cooking in food industry			
CO4	Explain the concept and applications of hurdle technology and image processing technology in food processing industries		
M4.01	Describe the principles and applications of Hurdle technology.	5	Understanding
M4.02	Explain the principles and applications of Nanotechnology	5	Understanding
M4.03	Explain the concepts and techniques involved in image processing	4	Understanding
	Series Test–II	1	
Contents: Hurdle technology - Concept of hurdle technology and its application, Nanotechnology - Concept of Nanotechnology and its application, Image processing technology-concepts and techniques involved in image processing			

Text / Reference

T/R	Book Title/Author
R1	James G Brennan, Alistair S Grandison (Ed.), Food processing Handbook, Wiley – VCH, 2011 2. P G Smith, Introduction to Food Process Engineering, Springer, 2011 3. Zeki Berk, Food process engineering and technology, Elsevier, 2013
R2	P G Smith, Introduction to Food Process Engineering, Springer, 2011

R3	Zeki Berk, Food process engineering and technology, Elsevier, 2013
R4	Geankoplis, C.J. “Transport Processes and Separation Process Principles”, 4th Edition, Prentice Hall, 2003.
R5	McCabe W.L., Smith J.C. “Unit Operations in Chemical Engineering”, 7th Edition, McGraw – Hill Int., 2001

Online Resources

Sl.No	Website Link
1	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7353510/
2	https://www.intechopen.com/chapters/38363
3	https://www.carbotecnia.info/learning-center/filtration-methods/what-is-nanofiltration/?lang=en
4	https://puretecwater.com/reverse-osmosis/what-is-reverse-osmosis
5	https://crystalquest.com/pages/what-is-ultrafiltration